MIL-S-20428
27 November 1951

MILITARY SPECIFICATION

SHELL, STEEL, MANUFACTURED FROM BAR STOCK

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-30-4D
4 December 1945

This specification consists of this cover sheet, Specification 50-30-4d, dated 4 December 1945, and Amendment 1, dated 22 December 1948, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-30-4D
4 December 1945
Superseding
No. 50-30-4C
11 December 1939

SHELL, STEEL, MANUFACTURED FROM BAR STOCK

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawings pertaining to the shell under contract, and all drawings and specifications listed thereon, shall form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of the invitation for bids.

B. CLASSES AND TYPES.

B-1. This specification covers minor caliber point-fuzed and base-fuzed, high-explosive, chemical, and practice shell machined directly from bar stock.

C. MATERIAL AND WORKMANSHIP.

C-1. All material and parts shall be in accordance with applicable drawings and specifications.

C-2. Steel bars. The shell shall be manufactured from cold-drawn steel bars complying with the grade and specifications shown on the drawing for the shell.

C-3. The machining and treatment of the shell shall be uniform for each lot so that samples selected for test will be truly representative of all the shell in the lot.

C-4. The interior of all shell shall be smooth and free from sharp edges, burrs, fins, etc.

C-5. All shell shall be sound and free from cracks and porosity. (See par. E-4).

C-6. The rotating band shall be pressed into the band seat so as to completely fill it and fit tight. It shall be applied cold in such a manner as not to distort the wall of the shell.

C-7. The machined rotating band shall be carefully protected against nicks, burrs, and other similar damage.

C-8. Cleaning prior to painting. Projectile surfaces to be painted shall be cleaned by one of the methods covered by Specification AXS-1245. These cleaned surfaces shall not be recontaminated prior to painting. At the time of painting these surfaces shall meet the various performance requirements of Specification AXS-1245.

C-9. The interior and exterior coatings specified on the drawing for the shell under contract shall be applied in such a manner as to insure an even distribution and a thorough coating. The shell shall be stenciled or marked as prescribed on the drawing.

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D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. The shell shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Rotating bands shall be properly seated. (See par. F-3).

E-3. The material of the rotating band shall show in test specimens taken from bands removed from the shell, the following physical properties:

	Copper	Gilding Metal
Tensile strength, lb. per sq. in., min.	30,000	32,000
Elongation percent in 1 inch, min.	15	15
Bend test degrees	180	180

Rotating bands manufactured from metal powder compact shall not be required to meet the above physical requirements after removal from the shell.

E-4. Shell shall be free from definite indications or cracks as defined below:

E-4a. Any cracks or indications other than tool marks on the shell which do not run parallel to the axis.

E-4b. Any cracks or indications running parallel with the axis which appear in the band seat, or the crimping groove, or which have a depth greater than 0.13 inch as determined by their extension into the non-cylindrical portion of the shell.

E-4c. Any cracks or indications parallel to the axis, which although not continuous, are definitely deep seated.

E-4d. Shell containing indications, not defined above, will be considered acceptable.

F. METHODS OF SAMPLING, INSPECTION, AND TEST.

F-1. Inspection. Inspection shall be as prescribed in U. S. Army Specification No. 50-0-1, General Specifications for Ammunition Except Small Arms Ammunition, and Federal Specification QQ-M-151, Metals, General Specifications for the Inspection of.

F-1a. Cracks and other defects. All shell shall be examined for defects as outlined in paragraph E-4 under a three power minimum magnifying glass with sufficient illumination to aid in detection of defects. Acceptability of borderline material shall be determined by comparison with samples furnished or approved by the Chief of Ordnance.

F-2. Lots.

F-2a. For high explosive and incendiary shell. The maximum size of lot shall be all shell made from the bars comprising one heat of steel only. The bars in tag ends of miscellaneous heats may be grouped for machining into miscellaneous lots not exceeding 10,000 shell per lot

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F-2b. For practice shell. The maximum size of lot shall be 100,000 shell with no restriction as to segregation of the bars comprising different heats of steel.

F-2c. A smaller quantity may constitute a lot at the discretion of the inspector.

F-2d. The first lot manufactured under any contract shall not exceed 10,000 shell, except when the new contract immediately follows the manufacture of satisfactory shell of the same character.

F-3. Rotating band tests.

F-3a. Detailed inspection.

F-3a(1). Method of inspection. One band per 5,000 shell from each banding machine shall be removed for examination of the seating. Prior to removal, the diameter of the finished band shall be measured on at least three diameters 60 degrees apart and on at least two points along the length of the cylindrical portion, near the front and near the rear (a minimum of six diametral measurements). These points of measurement shall be marked on the bands as well as the orientation of the band on the shell. After removal, the thickness of the band shall be measured at each of the marked points (twelve measurements) and the diameter of the band seat shall be measured opposite those points. Add to the diameters of the band seat at each point, the corresponding sum of the two thicknesses of the band and subtract the result from the corresponding exterior diameter of the band previously determined. The result will indicate the total diametral clearance between the band and its seat. If the band is well seated a negative clearance or interference may be indicated on some of the measurements or even on the average of all measurements. In case the indicated clearance thus determined is greater than .006 inch on any one diameter, either front or rear, the band shall be removed from ten additional shell from the group represented. If a clearance greater than .006 inch on any one diameter is indicated on any one of these additional shell the entire group represented shall be rejected subject to rebanding and retest. Bands which are removed for measurement except as indicated below will be weighed and the weights recorded and compared as to uniformity of successive groups and with prescribed weight shown on the drawing. When irregular band profile interferes with ease of measurement, the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The diameter of the machined band shall exceed the bourrelet diameter. These bands need not be weighed.

F-3a(2). Character of seating. Each band removed in accordance with paragraphs F-3a(1), F-3b and F-3c shall be examined for character of seating as indicated by imprints on interior of band corresponding to tool marks on the band seat, and to determine whether corners of the band seat have been properly filled. (See par. F-8a).

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F-3b. From each 5,000 shell a sufficient number of unmachined bands, or bands from which a light surface cut (.02 inch maximum) has been taken, shall be removed to provide one specimen to be subjected to each of the physical tests prescribed in paragraph E-2. Whenever practicable, bands removed for the seating test may be used. The bands shall be carefully removed from the shell and carefully flattened without hammering. Each test specimen shall be cut along the direction of the flattened length and prepared in such manner as to hold to the minimum any additional cold working. Tension test specimens shall be machined to the form and dimensions prescribed in U. S. Army Specification No. 50-27-1.

F-3c. In the bend test, each band or section thereof, of full width and thickness except as reduced by authorized surface cut, shall be bent cold by pressure, without hammering, in the direction of curvature of the band, until it tightly inclosed 180 degrees of a rod equal in diameter to the thickness of the bend test specimen (within $\pm$ .01 inch).

F-3d. If any specimen fails to meet the tensile or elongation requirements of the physical test, or if any band cracks in the bend test, a retest may be made at the request of the contractor. Double the number of samples shall be used for the retest and if failure occurs in the retest, the group of shell represented by the sample shall be rejected subject to rebanding and resubmission for test.

F-4. Proving ground test.

F-4a. Twenty-five sample shell shall be selected from each lot for the proving ground test.

F-4b. At the proving ground, the sample shell shall be loaded to service weight using an inert filler of the same weight and density as that of the explosive charge being simulated, and fired through paper screens placed in front of the muzzle of the gun at the approximate distance specified opposite the shell in the following table:

Projectile	Approximate Distance from Muzzle to Screen Feet	Gun
20MM H.E.I., Incendiary and Practice Shell	30 and 300	Hispano
37MM Shell, H.E., M54	20 and 60	M1A2
37MM Shell, Practice, M55A1	20 and 60	M1A2
37MM Shell, H.E., M63	25 and 50	M1A2
37MM Shell, T.P., M92	20 and 60	M1916
37 MM Shell, T.P., M94	20 and 60	1.457 S.C.

The shell shall be fired from a gun for which the shell is standard, using a propellent charge that will give twelve percent excess chamber pressure over the rated maximum pressure for the gun.

F-4c. Examine the paper screen to determine that the shell are not breaking up and that the rotating band is remaining in position.

F-4d. Proving ground test is not a condition for acceptance, but the results will be scrutinized carefully by the Ordnance Department to determine quality of the ammunition produced. Failure in any phase of the proving

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ground tests will result in reinspection of the lot by the Ordnance Department. If the reinspection reveals that latent defects were responsible for the failure the lot will be rejected. Improper seating of the rotating band will be considered to be a latent defect. (See par. F-3).

G. PACKAGING, PACKING, AND MARKING

G-1. Packing.

G-1a. Shell shall be packed in commercial type containers so constructed as to insure acceptance by common or other carrier, for safe transportation at the lowest rate to the point of delivery.

G-1b. Data cards. A data card, 5 x 8 inches, made of commercial manila tag board approximately 200 pounds ream weight, shall be prepared for each lot of accepted shell in accordance with the form specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be made in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind	Ammunition Lot No. . . . .
Quantity in lot.	. . . . .
Drawing No . . . . Rev . .	Spec No. . . . . Rev . . .
Manufactured by.	Date . . . . .
Contract No	Order No, ...
Bar Stock Manufacturer . . .	Heat No . . . . .
Packed.	. . . . . in a box.
Remarks	. . . . .
	Certified to by . . . . Inspector
	(Print here name of inspector, district, arsenal, or loading plant, and subdistrict where applicable).

G-2. Marking. The marking for shipment shall be in accordance with U. S. Army Specification No. 100-2, Standard Specification for Marking Shipments by Contractors. Lot number of contents shall be marked plainly on each container.

H. NOTES

H-1. This specification covers point-fuzed and base-fuzed, high explosive, chemical and practice shell manufactured directly from bar stock, except armor-piercing shell and shell for smooth-bore weapons.

H-2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE. This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

NOTE. Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington 25, D. C.

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No. 50-30-4D

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U. S. ARMY
SPECIFICATION

No. 50-30-4D
Amendment 1
22 December 1948

SHELL, STEEL, MANUFACTURED FROM BAR STOCK

The following changes shall be made in Specification No. 50-30-4D, dated 4 December 1945:

Page 1, paragraph C-8. Delete paragraph in its entirety and substitute:
"C-8. Cleaning Prior to Painting.-Projectile surfaces to be painted shall be cleaned and prepared in accordance with grade 1, Specification JAN-C-490. These surfaces shall not be recontaminated prior to painting. After painting, these surfaces shall meet the various performance requirements of Specification JAN-C-490."

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